Coprime Pell products and rational projection for Diophantine consistency

Research notes for ProveIt

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Abstract

This iteration replaces several independent square-root constraints by coprime products, and makes an existing rational Pell root carry the congruences formerly enforced by separate coordinates. The resulting polynomials retain explicit fixed coefficients and ordinary two-direction proofs of their zero-existence equivalence to the calibrated source.

The completed integer constructions include eleven variables at degree 688 and 126 arithmetic operations, compared with the previous 2240/206; a lower-operation eleven-variable alternative has degree 1632 and 114 operations. A thirteen-integer companion reaches degree 274 with 129 operations, versus the recent 536/135 construction. Twelve- and fourteen-integer tradeoffs have degree/operation pairs 500/148 and 108/119, respectively. A rational first-root projection gives fifteen integer variables at degree 56 and 110 operations, compared with the recent 172/123. A distinct coprime fusion gives ten positive variables at degree 266 and 96 operations, improving both late 340/103 and 368/98 alternatives.

A separate reusable result represents positive exponentiation with nine natural auxiliary variables, and shares the same frame among several power graphs. Its generic tiling composition is distinguished from a calibrated consistency equation. No global minimality, consistency proof or new end-to-end proof-assistant certificate is claimed.

New consistency equation	Unknowns	Exact degree	Operations
Selective coprime fusion over $\mathbb{Z}$	11	688	126
Direct selective fusion over $\mathbb{Z}$	11	728	122
Complete coprime fusion over $\mathbb{Z}$	11	1632	114
Four-root monic carrier over $\mathbb{Z}$	12	500	148
Three-root monic carrier over $\mathbb{Z}$	13	274	129
Two-root monic carrier over $\mathbb{Z}$	14	108	119
First-root monic carrier over $\mathbb{Z}$	15	56	110
Coprime Pell fusion over $\mathbb{Z}_{>0}$	10	266	96

The twelve- and fourteen-integer rows trade a small operation-count increase against substantially lower degree relative to the latest cheapest same-variable comparators. They are not asserted to dominate those points in every metric. The standalone power-graph result is not included in this table of calibrated consistency equations.

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1 What is optimized and what is proved

The object sought is a fixed polynomial with integer coefficients, not a family with an uninstantiated machine parameter. Its quantified variables have one explicitly stated domain. A polynomial abbreviation and a shared arithmetic register introduce no extra variables; conversely, a quantified witness hidden inside a special-function graph must be counted.

We compare the number of quantified variables, exact total degree, and live binary arithmetic operations. Fixed integer constants are coefficients, not additional unknowns. Addition, subtraction and

multiplication each cost one operation, including multiplication by a small constant. This convention measures a shared straight-line circuit, not expanded monomial count or bit complexity. Large fixed coefficients remain an important separate cost. There is no assertion that these measures admit a common global minimum.

The new results are ordinary mathematical reductions. Their concluding consistency statements compose these reductions with explicitly identified, pinned arithmetic-kernel, initialized-machine and syntactic-refutation calibration theorems. The accompanying programs check literal circuits, identities, degree certificates and fixed coefficient payloads. They do not replace the unbounded mathematical proofs, and do not constitute a new end-to-end Lean, Rocq or encoded object-language ZFC certificate.

In particular, the assertion is an equivalence

$$\mathsf{ZFC} \vdash (\neg\exists \mathbf{x} \in D^n \ P(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}),$$

using the stated ordinary calibration. It is not a proof of either side. Here $D = \mathbb{Z}$, $\mathbb{Z}_{>0}$ or $\mathbb{N}$ is specified separately for each construction, and $\mathbb{N}$ includes zero.

2 Coprime products and monic congruence carriers

The elementary observation driving this iteration is stronger when used in the design of witnesses rather than applied after a polynomial has already been built.

Lemma 2.1 (Coprime square extraction)**.** *Let $a_1, \dots, a_m$ be positive, pairwise coprime integers. Their product is a square if and only if each a_i is a square. If a, b, d are positive integers, ab/d is a rational square and $\gcd(a, bd) = 1$, then a is a square.*

Proof. For every prime dividing a_i, the valuation of the product is exactly the valuation of a_i. It must be even. The rational assertion uses the same valuation computation, with the denominator contributing zero at these primes. Positivity fixes the sign ambiguity. □

For example, a and $1 + adu^2$ are automatically coprime. Imposing that their product is a square therefore recovers two square conditions from one. This is not a universal equivalence between arbitrary two square conditions and one product condition: the coprimality and sign proofs are essential. It also changes the original witness problem. Completeness must show that the newly constrained Pell ordinate can actually be chosen.

Lemma 2.2 (A useful two-root criterion)**.** *Let $a, b > 0$ be rational numbers with $a \neq b$. If a signed sum of their square roots is rational, both square roots are rational.*

Proof. The sum cannot be zero. If $z = \epsilon\sqrt{a} + \eta\sqrt{b} \neq 0$, then $2z\epsilon\sqrt{a} = z^2 + a - b$, and similarly for the other root. □

Neither statement can be used without its hypotheses. The integer constructions below prove positivity before splitting products. The positive-variable construction proves the two radicands unequal by a congruence obstruction, before making any parity or transport deduction.

Lemma 2.3 (A monic carrier for two congruences)**.** *Let M, H be nonzero coprime integers, and let n, K, v, v_0 be integers. Set $U = MH$ and $Z = Un + MK + H(v - v_0)$. If Z/U is a root of any monic polynomial with integer coefficients, then*

$$M \mid v - v_0, \qquad H \mid K.$$

Proof. A rational algebraic integer is an integer, so $U \mid Z$. Reducing modulo M and H and using coprimality gives the two conclusions. □

This observation changes the order of the proof. A monic radical norm can force the scale and projection congruences before any individual radical has been shown rational. Its norm equation alone makes the total rational center integral. The restored scale can then justify the bounds or prime-valuation arguments needed to recover each separate Pell root. This avoids a circular appeal to the very scale that the equation is supposed to enforce.

3 Pinned comparison and attribution

The comparison is pinned to ProveIt commit `84e16e43092614bae262ca59651cb9d77e31a255`. The recent source changes matter: they already improve several endpoints relative to the previous delivery. In particular the incoming compact-proof packet has a fifteen-integer degree-172 circuit with 123 operations, which supersedes the previous degree-176, 125-operation comparison. We use the stronger available comparator below.

Domain and construction	Variables	Degree	Operations
Previous eleven-integer delivery	11	2240	206
Recent scalar carrier	12	1200	145
Recent scalar carrier, integral variant	12	1456	141
Recent thirteen-integer norm	13	536	135
Recent guard-reduced integer form	14	312	117
Incoming compact-proof integer form	15	172	123
Recent direct-base positive form	10	340	105
Recent cheaper positive form	10	368	99
Recent nine-positive circuit	9	1191570	405

These rows do not exhaust the Pareto frontier. The repository also has lower-variable, much higher-degree equations; low-operation formulas with more variables; and direct quartic compiler constructions. At the initial comparison pin the direct projected quartic has 604 unrestricted integer variables. A bounded late review through commit `56b4d15d128b` finds a completed 601-integer projected quartic. The subsequent 73-predicate source theorem does not yet supply a measured smaller projected quartic. Such expanded quartic sizes are not comparable to short circuit counts without also accounting for expanded support and coefficient height.

The scalar intersective carrier, signed absolute Pell folding, full-residue interpolation and centered budget are inherited ideas and are credited as such. The new contribution is their use with deliberately coprime square products and, separately, with a rational first-root projection carrying two congruences. We do not claim historical priority for the elementary square-product lemma or a literature-wide minimum.

The archive's baseline comparison records exact source paths and hashes. All original source files and older candidate polynomials remain unchanged. Original-label source34 and compact44 coefficients are kept in separate coherent sets. In particular no original-label initialization coefficient is identified with the repaired BFS-renamed initialization coefficient.

A late transport expansion in upstream commit `49379c31d969` reduces the two ten-positive comparator circuits to 103/340 and 98/368 (operations/degree). Applying that exact identity to the new fused polynomial gives 96/266, without changing its coefficients, degree or zero set. This late simplification is credited to the upstream construction. The initial comparison table remains a record of the frozen starting pin; the final positive comparison uses the late figures.

4 Eleven unrestricted integers: selective coprime fusion

The principal new equation has **eleven unrestricted integer unknowns, exact degree** 688**, and** 126 **live binary arithmetic operations**. Its direct sibling has degree 728 and 122 operations. The preceding eleven-integer endpoint had degree 2240 and 206 operations. A more aggressive fusion gives a lower-cost alternative of degree 1632 and 114 operations. These are comparisons of complete displayed circuits, not global optimality claims. The unbounded native/refutation calibration remains inherited; no new formal kernel certificate is claimed.

4.1 The complete polynomial

The independent coordinates are

$$(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}.$$

Use the six fixed integers $\beta, K_0, M_C, M_F, A_c, B_c$ specified by the retained source-34 coefficient recipe, with M_F already shifted. Here $A_c = 2^{1113}$ and $B_c = 2^{666104}$; neither is a variable.

All subsequent symbols are polynomial abbreviations:

$$\begin{aligned}
J &= 6j+1, & q &= \beta J + 1, & W &= 12w+4,\\
X &= qW, & Y &= (2v+1)q^3, & E &= XY,\\
a &= E+Y, & H &= 4a+3, & \Delta &= a^2+H,\\
F &= (K_0+W)C - z(q-1), & & & &\\
R &= (q(q-F)-C)(q^2-1) + (M_C+qM_F)J, & & & &\\
k &= R+1+hE, & \eta &= c - Yk, & &\\
D_{\mathrm{err}} &= (A_cF-q)^2 + (B_cC-q)^2, & & & &\\
B &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, & & & &\\
L &= XY^2k, & D_t &= 1 + L(L+k), & &\\
D &= 3B-4, & D_g &= 1 + 4DD_ts^2, & P &= D_tD_g,\\
I &= 2src^2, & D_f &= 1 + \Delta I^2, & Q &= \Delta^2 I^2,\\
\delta &= Q-1, & V &= D_f(c - R + 4IT) - c, & &\\
C_y &= QV^2 - 1, & D_a &= \delta C_y, & &\\
K &= X(3c-2X)+2, & S &= IH^5, & z_c &= HT+K,\\
A_* &= (z_c^2+3H^2)(z_c^3-19H^3), & & & &\\
Z &= S(n+(2Q-1)V) - A_*, & & & &\\
u &= S^2D_f, & b &= 4S^2D_a, & e &= 16S^2P.
\end{aligned}$$

The promised single polynomial equation is

$$\boxed{\left[(Z^2-u-b-e)^2 - 4(ub+ue+be)\right]^2 - 64ubeZ^2 = 0.} \tag{1}$$

There are no radicals, divisions, variable exponents or implicit witnesses in this display. Removing $(2Q-1)V$ from Z gives the direct 728/122 variant by an integral translation of n.

4.2 Why the fusion is valid

The polynomial in (1) is the product of the eight independent conjugates of $Z - \sqrt{u} - \sqrt{b} - \sqrt{e}$. The former five-root construction is reduced in two different ways: the main root is recovered arithmetically

rather than included in the norm, and two other square conditions are fused using

$$\gcd(D_t, D_g) = \gcd(D_t, 1 + 4DD_t s^2) = 1.$$

Thus $D_t D_g$ is a positive square exactly when both coprime positive factors are squares. This is not an assumption that arbitrary products of squares can be reversed.

For every integer assignment $H \neq 0$, $\Delta > 1$, $D_f, D_t \geq 1$. If $I = 0$, the equation specializes to $A_*^8 \neq 0$, because

$$A_* = H^5 \pi(T + K/H), \qquad \pi(x) = (x^2 + 3)(x^3 - 19)$$

has no rational zero. Hence $I, s, r, c \neq 0$ at any zero. Then $|I| \geq c^2$, $D_f > |c| + 1$, $|V| \geq 1$, and $D_a > 0$. If $D_g < 0$, its fused radical would be the only imaginary term in a vanishing factor, an impossibility. Therefore $D_g > 0$, and $B \geq 2$. The centered budget supplies the common strict signs of $q, F, C, q - F - C$ and $|c/k - Y| < 1/2$.

All three radicands D_f, D_a, P are odd positive integers. In a rational signed sum with weights $1, 2, 4$, each irrational squareclass has a unique least 2-adic weight and cannot cancel. Thus all three roots are integers. Coprimality recovers separate square roots of D_t, D_g. The auxiliary quotient $y = \sqrt{D_a}/\delta$ is integral by the retained adjacent-Pell descent. The factor identity now implies $S \mid A_*$. Monicity and prime-valuation comparison give $H \mid K$. Since

$$K = HXc - 2\{X(X + 2ac) - 1\}, \qquad H \text{ odd},$$

the signed quadratic main projector is recovered as well.

The missing main Pell norm must not be assumed in this argument. From the strong and auxiliary norms, absolute-value Pell folding gives $|c| = \psi_{|a+2|}(p)$ for some $p \geq 1$ before projection. The retained centered signed-rank proof then yields $p = R = 2t + 1$, $t \geq 24$, and the requisite ratio bound. The signed whole-product estimate yields $\operatorname{sign}(c)X = 2^R$; the original centered half-binomial argument restores positive orientation and the native interval. Fresh private witnesses then restore the exact old signed kernel and its finite syntactic refutation endpoint. The complete proof records each dependency and its order in `integer/PROOF.md`.

Conversely, a finite refutation supplies the same canonical outer tuple, with $B > 64$ and $D_t = \tau^2 > 0$. Since $(3B - 4)D_t \equiv 2 \pmod 3$, ordinary Pell existence and a squared unit supply $s, g > 0$ with $D_g = g^2$. The retained full-residue interpolation theorem then chooses one fixed strong pair I, f before any residue for T is requested. The intersective polynomial π has a root modulo every integer; choose T so that $I \mid \pi(T + K/H)$, and finally set

$$n = A_*/S + f + 2\delta y + 4\tau g - (2Q - 1)V.$$

This makes a conjugate factor zero. In particular, the strong witness is not varied after selecting the intersective residue.

4.3 Exact degree and independently checked arithmetic

The relevant degrees are

$$\deg(S, D_f, Q, V, D_a, D_t, D_g, P) = (34, 20, 32, 25, 114, 34, 60, 94).$$

Thus $\deg(u, b, e) = (88, 182, 162)$. Put $C_* = S(2Q - 1)V$ and $s_* = Sn - A_*$. The exact identity

$$b - C_*^2 = S^2(-V^2 - 4Q + 4)$$

gives $\deg(Z^2-b)\le 126$. Rewriting the norm with $\omega=Z^2-b$, its inner bracket has unique leading term $-4be$ of degree 344; the remaining outer correction has degree 614. Hence the output has degree 688 and leading form $16b_0^2e_0^2$. For the authentic fixed tuple its coefficient

$$[j^{376}w^{84}v^{96}h^{12}s^{32}r^{28}c^{56}T^4]E_8=2^{780}3^{462}\beta^{376}>0$$

proves that this bound is attained. No generic-coefficient assumption or random sampling is used.

The portable checker authenticates all six numeral payloads, every live arithmetic register, all 35 coefficients of the free three-root norm, all degree-leading inductions and the centered cancellation cut. Both ordinary and optimized Python runs pass. Counting every live binary addition, subtraction and multiplication, including squares and scalar products, gives exactly 126 operations; leaves and reuse are free.

4.4 A lower-operation alternative

Replace I by

$$I=2src^2D_tD_gD_m(a+2),\qquad D_m=1+\Delta c^2,$$

recompute the dependent definitions, and set $Z=Sn-A_*$. The equation

$$\boxed{\delta Z^2-S^2D_fD_tD_gC_y=0}$$

has degree 1632 and 114 operations, still in eleven integers. It is a polynomial despite its derivation from a rational auxiliary root. Now D_f,D_t,D_g,D_a are pairwise coprime: the new factors in I give the first four coprimalities, while

$$D_a\equiv(a+2)^2D_m\pmod{D_f}$$

gives the last one. Squaring $\delta Z/S$ recovers the product of these four positive coprime integers; each is a square. The same adjacent-Pell descent and scalar projection finish soundness. All additional factors in I are fixed before applying the same full-residue interpolation lemma, so completeness introduces no stronger unproved interpolation assumption. The detailed proof and complete circuit accompany the package.

5 A first-root carrier: fifteen integers, degree 56

The principal new equation has **15 unrestricted integer unknowns, exact degree 56, and 110 live binary arithmetic operations**. It improves both degree and evaluator cost over the inspected 15-variable degree-170 and degree-172 alternatives. The construction is not a generic variable renaming: it makes the first Pell root carry two coprime congruences, replacing the separate main-root coordinate and a large radical norm.

The unknowns are

$$j,C,z,h,r,v,w,T,c,n,f,y,p_1,p_2,p_3\in\mathbb{Z}.$$

Use the same six fixed source34 coefficients as in the retained calibrated source, with M_F already shifted.

All expressions below are abbreviations, not additional unknowns:

$$\begin{aligned}
J &= 6j+1, & q &= \beta J+1,\\
W &= 12w+4, & X &= qW,\\
Y &= qv, & F &= (K_0+W)C - z(q-1),\\
E &= XY, & a &= E+Y,\\
H &= 4a+3, & \Delta &= a^2+H,\\
k &= R+1+hE, & \eta &= c-Yk,\\
L &= EYk, & I &= (2r+1)c^2,\\
Q &= \Delta(f^2-1), & V &= f^2(cT-R)-c,\\
M &= 2q^2, & U &= MH,\\
K_* &= 2X^2-3Xc-2, & Z &= Un+MK_*+H(v-q^2).
\end{aligned}$$

Here the register used above is

$$R = (q(q-F)-C)(q^2-1) + (M_C + qM_F)J.$$

Put

$$\begin{aligned}
D &= (A_cF-q)^2 + (B_cC-q)^2,\\
B_{\text{sign}} &= k^2(q^2-D) - 4q^2\eta^2,\\
P_{\text{pos}} &= 1+p_1^2+p_2^2+p_3(p_3+1),\\
A_1 &= Z^2 + (2Z-Uk)UL - U^2,\\
A_2 &= f^2 - \Delta I^2 - 1,\\
A_3 &= V^2 - (Q-1)y(2V+y) - 1,\\
A_4 &= B_{\text{sign}} - P_{\text{pos}}.
\end{aligned}$$

The complete equation is

$$\boxed{P_{15} = A_1^2 + A_2^2 + A_3^2 + A_4^2 = 0.} \tag{2}$$

Its literal DAG is `lowdegree/equation.json`.

Why the compression is sound. Unconditionally, q is even, $q \equiv 2 \pmod 3$, H is odd, and $H \equiv 3 \pmod q$. Hence $\gcd(H, M) = 1$ and $U \neq 0$. The first row makes $t = Z/U$ a rational root of the monic integer polynomial $t^2 + (2t-k)L - 1$, so $t \in \mathbb{Z}$. Coprime CRT gives

$$v = q^2(2v_0+1), \qquad H \mid K_*.$$

Thus the cubic scale is recovered *before* the signed Pell bootstrap. For $K = X(X+2ac)-1$, the identity $2K - K_* = HXc$ implies $H \mid K$. The budget row supplies the common native signs and $|\eta/k| < 1/2$. The strong and auxiliary rows recover the omitted main Pell square by the ordinary folded-index lemma. Normalized rank then gives its index R. The retained whole-product estimate $|H| > 2^{2R+2}|X|^2$ turns $H \mid K$ into the exact signed power relation, and restores an integral main coordinate $\gamma = (d - X - ac)/H$. This yields a literal zero of the pinned normalized 17-variable source. The proof does not assume positivity of η before the source's centered half-binomial argument recovers it.

Conversely, every zero of that source has even q, hence even old J_3. Set $j = J_3/2$, $C = q - F - \alpha$, $v = q^2(2v_0+1)$, and $n = t_{\text{old}} - K_*/H - v_0$. The old main row proves $H \mid K_*$; all the displayed coordinates are integers and all four new rows vanish. The complete signed proof and maps are in `lowdegree/PROOF.md`.

Exact, not merely syntactic, size. The row degrees are $28, 14, 18, 16$. The actual fixed-coefficient polynomial has

$$[j^{30}w^{10}v^{12}h^4]P_{15} = 2^{62}3^{40}\beta^{30} > 0.$$

Thus its exact degree is 56. Its 110 live gates are 63 multiplications, 24 additions and 23 subtractions. Independent normal and optimized-Python checks verify all register cuts, coefficients, liveness, the leading form, and rejection of corrupted gates.

6 A double-centered companion: fourteen integers, degree 108

The second equation removes the explicit f, leaving **14 unrestricted integers, exact degree 108, and 119 live operations**. Keep the preceding abbreviations except replace f^2 everywhere by

$$D_f = 1 + \Delta I^2, \qquad Q = \Delta(D_f - 1), \qquad V = D_f(cT - R) - c.$$

Define the two doubled root centers and errors

$$\begin{aligned} A &= 2L + k, & B &= 2(a+2)I, \\ a_* &= 4 - k^2, & b_* &= 4(1 - I^2), \\ D_0 &= Z^2 + 2BUZ + U^2(a_* - b_*), \\ E_0 &= Z + U(A + B), \\ B_0 &= A[Z^2 + 2BUZ - b_*U^2] - a_*U[Z + UB], \\ N &= D_0^2 + 4UE_0B_0. \end{aligned}$$

Now $A^2 + a_* = 4D_\tau$, $B^2 + b_* = 4D_f$, where $D_\tau = 1 + L(L + k)$. The complete equation is

$$\boxed{P_{14} = N^2 + [V^2 - (Q-1)y(2V + y) - 1]^2 + [B_{\text{sign}} - P_{\text{pos}}]^2 = 0.} \tag{3}$$

The shorter evaluator uses the identical unexpanded norm

$$N = \{E_0^2 - 4U^2(D_\tau + D_f)\}^2 - 64U^4D_\tau D_f.$$

The centered expression above supplies the degree certificate, rather than requiring its longer evaluation schedule. The exact two-root norm identity makes $Z/U + A + B$ a rational root of a monic quartic over $\mathbb{Z}$. Hence CRT again recovers the scale first, without assuming radical rationality or excluding $v = 0$ by hand. The guard and recovered scale give $\sqrt{D_f} > 2\sqrt{D_\tau}$, so the nonzero rational signed root sum forces both roots rational and therefore integral. This restores the 15-variable equation by selecting $f = \sqrt{D_f}$ and an integral new n. Conversely every 15-variable zero gives a zero here by shifting n to the sum of its doubled integer roots minus $A + B$ and the CRT offsets.

The fully centered norm has degree 54, not the larger bound obtained by expanding an uncentered norm syntactically. In the final polynomial,

$$[j^{56}w^{18}v^{22}h^4r^2c^4n^2]P_{14} = 2^{130}3^{74}\beta^{56} > 0.$$

Its 119 live gates are 69 multiplications, 30 additions and 20 subtractions. The literal equation, full proof, and separate independent review are under `lowdegree/fourteen/`.

Common semantic boundary. Both reductions use the unchanged authenticated source34 coefficient tuple and the retained ordinary normalized-source, native compiler and finite-ZFC- refutation calibration. They do not prove consistency, claim global minimality, or supply a new end-to-end proof-assistant or encoded internal-ZFC certificate. Their two-way arithmetic maps are explicit, with no hidden existential registers or domain restrictions.

7 A triple-centered companion: thirteen integers, degree 274

A third endpoint removes the auxiliary coordinate y as well as f. It has **13 unrestricted integer variables, exact degree 274 and 129 live operations**, improving both degree and evaluator cost over the inspected 13-variable degree 536/135-gate equation.

Keep the shared definitions with $D_f = 1 + \Delta I^2$, $Q = \Delta(D_f - 1)$, $V = D_f(cT - R) - c$, and $Z = Un + MK_* + H(v - q^2)$. Set

$$\begin{aligned} D_\tau &= 1 + L(L + k), & D_a &= (Q - 1)(QV^2 - 1), \\ A &= 2L + k, & B &= 6(a + 2)I, \\ C_* &= 9(2Q - 1)V, & Z_* &= Z + U(A + B + C_*), \\ s &= Z_*^2, & u &= 4U^2 D_\tau, \\ v_* &= 36U^2 D_f, & w_* &= 324U^2 D_a. \end{aligned}$$

Define

$$N = [(s + w_* - u - v_*)^2 - 4(sw_* + uv_*)]^2 - 64sw_*uv_*.$$

With $G = B_{\text{sign}} - P_{\text{pos}}$, the complete equation is

$$\boxed{P_{13} = N + G^2 + NG^2 = 0.}$$

The unit-product identity $(1 + N)(1 + G^2) = 1$ proves $N = G = 0$ without any assumption about the sign of N.

Root recovery and complete maps. The norm is monic in the rational total center Z_*/U, so integer-root monicity again licenses CRT before root extraction. Unconditionally,

$$D_\tau \equiv D_f \equiv D_a \equiv 1 \pmod 3.$$

The root weights $2, 6, 18$ therefore have distinct three-adic orders. In any nonrational squareclass, the least-order term cannot cancel the others modulo 3; independence of squareclasses forces all three roots to be integers. If $e^2 = D_a$, the rational adjacent-Pell root $e/(Q - 1)$ is integral by the ordinary denominator-descent lemma, explicitly proved in the component proof. Restoring

$$f_{15} = \sqrt{D_f}, \quad y_{15} = e/(Q - 1) - V, \quad n_{15} = \sqrt{D_\tau} - L - K_*/H - (v - q^2)/M$$

gives a literal 15-variable zero. Conversely every 15-variable zero gives an integral $e = (Q - 1)(V + y)$ and an integral shift of n making a norm factor zero. No new canonical-choice or strong-index tuning is required.

Why triple centering matters. The root-center defects are

$$4D_\tau - A^2 = 4 - k^2, \quad 36D_f - B^2 = 36(1 - I^2), \quad 324D_a - C_*^2 = 81(4 - 4Q - V^2).$$

They expose cancellations hidden by the compact direct evaluator. The independent generic expansion has 568 terms and exact weighted leading monomial $128ZU^7A^2BC_*^4$, of degree 242. Hence the signed norm itself has degree 242 and the unit-product finalizer has degree 274. The actual fixed-coefficient leading witness is

$$[j^{124}w^{41}v^{47}h^6z^4r^{17}c^{34}n]P_{13} = 2^{265}3^{174}\beta^{124} > 0.$$

The 129-gate direct evaluator has 76 multiplications, 31 additions and 22 subtractions. The longer Horner evaluator is retained solely as an exact identity/degree comparison. Full formulas, ordinary maps and independent checks are under `lowdegree/thirteen/`.

8 A four-root companion: twelve integers, degree 500

The final member removes p_1 as well, leaving **12 unrestricted integer variables, exact degree 500 and 148 live operations**. It improves both measures over the inspected degree 1000/180-gate twelve-variable endpoint. The latest degree-1200, 145-operation and degree-1456, 141-operation twelve-variable circuits remain cheaper, so the new 500/148 endpoint is a degree–cost tradeoff against those comparators. The guard is now a square radicand inside the norm itself, not an additional inequality or hidden witness.

Keep the 13-variable definitions and set

$$G_0 = B_{\text{sign}} - 1 - p_2^2 - p_3(p_3+1), \qquad Z_* = Z + U(A + B + C_*).$$

Define the four scaled radicands

$$a_r = 4U^2 D_\tau, \quad b_r = 36U^2 D_f, \quad c_r = 324U^2 D_a, \quad e_r = 729U^2 G_0,$$

and the direct four-root evaluator

$$\begin{aligned}
p &= a_r b_r, & q_* &= c_r e_r, & u_* &= a_r + b_r, & v_* &= c_r + e_r,\\
s_* &= Z_*^2 - u_* - v_*,\\
k_* &= s_*^2 - 4u_* v_* + 4(p + q_*),\\
\ell_* &= s_* + 2v_*, & m_* &= s_* + 2u_*,\\
E_* &= k_*^2 + 16[p\ell_*^2 - q_*(m_*^2 + 4p)],\\
O_* &= k_*\ell_* + 8q_* m_*.
\end{aligned}$$

The complete equation is

$$\boxed{P_{12} = E_*^2 - 64pO_*^2 = 0.}$$

It is exactly the product over all 16 signed sums of the four roots, with no finalizer needed.

Why the guard can join the roots. Monicity of the 16-sign norm first makes the rational total center integral, so the same coprime CRT recovers the cubic scale. The first three radicands remain units modulo 3. Their root weights $2, 6, 18$ have respective three-adic coefficient orders $0, 1, 2$; the fourth weight 27 has order at least 3 after the squarefree decomposition of nonzero G_0. Therefore no nonrational squareclass can cancel. If $G_0 = 0$, that root is simply omitted; a negative G_0 is excluded by the same squareclass argument, not assumed away. Thus $G_0 = g^2$ for an integer g and

$$B_{\text{sign}} = 1 + p_2^2 + p_3(p_3+1) + g^2 > 0.$$

Set $p_{1,15} = g$ and restore the other three integer roots exactly as in the 13-variable proof. The adjacent-Pell denominator lemma restores the auxiliary coordinate. Conversely every 15-variable zero gives this norm zero after an integral shift of n incorporating $27p_1$. Both maps apply to every zero, without extra canonical choices.

Exact degree. The triple-centered root defects are unchanged. The cleared 16 factors have degrees 47 eight times, 17 four times and 14 four times, so their product has exact degree 500. Its actual leading witness is

$$[j^{228} w^{76} v^{84} h^8 z^8 r^{32} c^{64}] P_{12} = 2^{488} 3^{332} \beta^{228} > 0.$$

The 148 live gates are 90 multiplications, 35 additions and 23 subtractions. The complete scalar definitions, ordinary proof and independent certificate are under `lowdegree/twelve/`.

9 Coprime Pell fusion: ten positive variables and degree 266

The following construction improves the pinned ten-positive-variable endpoints from $(103, 340)$ and $(98, 368)$, where the ordered pair records live binary arithmetic operations and exact total degree, to

$$\boxed{(96, 266)}.$$

The operation split is 57 multiplications, 20 additions and 19 subtractions. Multiplication by a constant counts as an operation. This is an ordinary unbounded arithmetic reduction relative to the retained source/native calibration, not a new Lean or encoded internal-ZFC certificate. No consistency or global minimality claim is made.

9.1 Complete polynomial

All ten variables

$$J, F, \alpha, i, T, x, Y, n, \eta, \zeta$$

range over strictly positive integers. The four fixed integer coefficients are the complete original compact44 tuple bound in `equation.json` and its coefficient payloads. In particular,

$$\beta \geq 31 \text{ is odd}, \qquad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta.$$

Here M_F already includes its source shift. The distinct repaired BFS44 tuple must not be mixed with this tuple; its K_0 is not the same integer. Define the following polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & X &= q + x, & \mu &= q^2 + 1,\\
A &= \mu Y(X+1) + 1, & \Delta &= A^2 - 1, & H &= 4A - 5,\\
k &= \eta + \zeta, & c &= kY + \eta, & V_0 &= X\mu^2 Y^2,\\
D &= V_0(V_0+1)k^2 + 1, & G &= (q-1)(q-F) + \alpha, & &\\
R &= G(q^2-1) + (M_C + qM_F)J, & e &= k - R - 1, & &\\
C &= q - F - \alpha, & B &= (qK_0 + X)C - qF, & &\\
E &= ic^2, & \mathcal{F} &= 1 + \Delta DE^2, & Q &= \Delta^2 DE^2,\\
\delta &= Q - 1, & b &= c(T+\alpha) - R, & V &= \mathcal{F}b - c,\\
D_y &= QV^2 - 1, & L &= HDE + b, & W &= 1 + q(q-1)L,\\
P &= q^4(q-1), & M &= P\mu^2 H, & K_+ &= X(3c - 2X) + 2.
\end{aligned}$$

Finally set

$$\begin{aligned}
Z &= P\mu^2\{H(n+V) + K_+\}\\
&\quad + H\{\mu^2[q(q-1)Y - q^3WB - (q-1)X] - Pe\},\\
a &= D\mathcal{F}, \qquad s = \delta Z^2, \qquad u = \delta M^2 a, \qquad v = M^2 D_y.
\end{aligned}$$

The single equation is

$$\boxed{\mathcal{N} = (s - v - u)^2 - 4uv = 0.} \tag{4}$$

There are no additional quantified abbreviation variables and no radicals or divisions in its evaluator. The complete circuit uses the stored names `Dtau`$= D$, `Df`$= \mathcal{F}$, and `I`$= E$. The latter is *not* the actual strong Pell ordinate introduced in the proof below.

The structural change is to make the first radicand D a factor of the strong squared ordinate. Since $\mathcal{F} \equiv 1 \pmod{D}$, a square product $D\mathcal{F}$ forces its two factors to be squares separately. This permits a two-radical quartic norm in place of the parent's three-radical octic norm.

9.2 Exact contraction of the literal evaluator

The first fused evaluator had 98 operations; the current 96-operation schedule is identically equal to it. Write $U = \mu^2$, $S = q^3$, $g = q(q-1)$ and $D_{\mathrm{low}} = (q-1)X$, so $P = Sg$ and $W = 1 + gL$. The literal evaluator uses

$$Z = PU\{H(n + V - LB) + K_+\} + H\{U(gY - SB - D_{\mathrm{low}}) - Pe\}.$$

Subtracting the earlier displayed expression gives $HULB(Sg - P) = 0$ universally. Delete the private gL, W, WB definitions and insert only LB and $(n + V) - LB$. Also compute $P = q^3[q(q-1)]$ using the existing $q(q-1)$ register, deleting the private q^4 register. This saves two operations; all remaining kernel and norm gates are unchanged. The final split is 57 multiplications, 20 additions and 19 subtractions. The abbreviation W remains useful in the proof but is not separately evaluated. The complete polynomial, its degree, and both positive witness maps are unchanged. This same contraction appears in upstream commit 49379c31d, yielding comparison endpoints 98/368 and 103/340; the fused endpoint is 96/266.

9.3 Root extraction and the implicit strong ordinate

The exact rational carrier identity is

$$\frac{Z}{M} = n + V + \frac{K_+}{H} + \frac{Y}{q^3} - \frac{WB}{q(q-1)} - \frac{X}{q^4} - \frac{e}{\mu^2}. \tag{5}$$

For every positive input, $q \geq 32$, $X > q$, $A > 4$, $c \geq 3$, $E \geq c^2$ and $D > 1$. The parent's elementary adjacent-radicand bounds, strengthened by the new factor D, give

$$\delta > c^2 - 1 > 0, \qquad \mathcal{F} > c + 2, \qquad \mathcal{F} < \delta.$$

If $b = 0$, then $V = -c$ and $c^2 < D_y/\delta < c^2 + 1$. If $b \neq 0$, then $|V| \geq \mathcal{F} - c > 1$. Thus a and D_y/δ are positive rational numbers before integrality is established.

They are unequal: $\delta a = D_y$ would reduce modulo $D > 1$ to $0 = -1$, since $D \mid Q$ and $D \mid a$. The elementary norm identity

$$(z^2 - r_1^2 - r_2^2)^2 - 4r_1^2 r_2^2 = \prod_{\epsilon,\nu \in \{-1,1\}} (z - \epsilon r_1 - \nu r_2)$$

applied to (4) therefore gives a nonzero rational signed sum

$$\frac{Z}{M} = \epsilon\sqrt{a} + \nu\sqrt{D_y/\delta}.$$

Both radicals are rational: squaring this relation and solving linearly for either radical proves the assertion, since $Z/M \neq 0$. As a is an integer, it is an integer square. Furthermore $\gcd(D, \mathcal{F}) = 1$, so unique factorization supplies positive integers

$$D = \tau^2, \qquad \mathcal{F} = f^2.$$

Define the actual strong ordinate

$$I_* = \tau E.$$

It is an integer even though it is not asserted to be a polynomial in the ten inputs. Then

$$f^2 - \Delta I_*^2 = 1, \qquad Q = (\Delta I_*)^2, \qquad c^2 \mid I_*, \qquad I_* \geq c.$$

The retained adjacent-coefficient rational-to-integral descent applied to $QV^2 - (Q-1)y^2 = 1$ proves that $y = \sqrt{D_y/\delta}$ is a positive integer. Consequently the carrier relation is

$$Z/M = \epsilon\tau f + \nu y. \tag{6}$$

None of these steps assumes parity of q, transport positivity, or $\mu^2 \mid e$.

9.4 Soundness with delayed denominator cancellation

Write $\chi_A(t), \psi_A(t)$ for the standard Pell sequences defined by

$$\chi_A(t) + \psi_A(t)\sqrt{A^2-1} = (A + \sqrt{A^2-1})^t.$$

The retained all-index strong/auxiliary folding lemma applies to A, c, I_*, f, V, y: its hypotheses include $A \geq 3$, $c \leq I_*$, $f > 2I_*$ and $V = -c \pmod f$. It gives $c = \psi_A(p)$ for an odd $p \geq 1$. The first Pell equation gives

$$k = 2\psi_{2V_0+1}(n_0), \qquad \tau = \chi_{2V_0+1}(n_0), \qquad n_0 \geq 1.$$

The case $n_0 = 1$ would give $c = 2Y + 1$ strictly between $\psi_A(1) = 1$ and $\psi_A(2) = 2A$. Hence

$$c > kY \geq k \geq 4(2V_0 + 1) > 8X\mu^2Y^2. \tag{7}$$

The unchanged mask estimate gives $R < c\alpha/4$. As in the parent this implies

$$b > 0, \quad V > 0, \quad |e/\mu^2| < b/2, \quad 0 < y - V < b/2.$$

The adjacent-root estimate uses $\mathcal{F} < \delta$. Also $c > 2X$, so $K_+ > 0$.

The new polynomial transport majorant is adequate because $f < AI_*$ and therefore $\tau f < ADE$. Since $D > X$ and $E \geq 1$,

$$\tau f + X/q^4 < (A+1)DE < HDE. \tag{8}$$

If $B \leq -1$, then $WB/[q(q-1)] < -L$. Solving (5)–(6) for n and using (8), the two $b/2$ bounds, $K_+/H > 0$ and $Y/q^3 > 0$ gives $n < HDE + b - L = 0$, a contradiction. Thus $B \geq 0$, and its definition forces $C > 0$. The unchanged mask bounds now give

$$3q < R < q^4 - 2.$$

The normalized strong-rank and signed auxiliary-index lemmas apply since $c^2 \mid I_*$, $c > 2R$, $c > 2p$, $V = -c \pmod f$ and $V = -R \pmod c$. They give $p = R$. Oddness of p gives R odd, and the original mask parity calculation gives q even and J odd.

Only at this point use

$$\gcd(\mu^2, q^4(q-1)H) = 1.$$

Indeed $H = -1 \pmod \mu$, $\gcd(q, \mu) = 1$, and evenness of q handles $q - 1$. Since (6) is integral, separating the single modulus μ^2 in (5) yields $\mu^2 \mid e$. The first Pell congruence and the unchanged strict comparison $2V_0 + 1 > A$ give $n_0 = (R+1)/2$. Put $R = 2r + 1$; then $r \geq 48$.

The retained zero-offset ratio lemma, uniform in $X \geq 32$, is

$$\xi < \frac{\psi_A(2r+1)}{\psi_{2X\mu^2Y^2+1}(r+1)} < \xi\left(1 + \frac{1}{Y(X+1)}\right)^{2r}, \qquad \xi = \frac{(X+1)^{2r}}{X^r}.$$

It gives $Y > X^r/3$ and $X^{r+1} < 3Y(X+1) < H$. From the integral center, $H \mid PK_+$. Since H is odd and

$$K_+ = HXc - 2K_2, \qquad K_2 = X[X + 2(A-2)c] - 1,$$

cancel only 2 and apply the universal projection lemma:

$$H \mid q^4(q-1)(X - 2^R)(2^RX + 1).$$

The whole product has absolute value less than $X^{r+1} < H$ when $X \geq q \geq 32$, $r \geq 48$, as in the direct-base parent. Thus $X = 2^R$. This is whole-product vanishing, not cancellation of q through H or a choice of a square-root branch separately at each prime. Projection now also makes K_+/H integral.

Removing the integral terms from (5) leaves $Y/q^3 - WB/[q(q-1)] - X/q^4$ integral. Clear the denominator and reduce modulo q to obtain $q \mid X$. Hence $q = 2^\kappa$; since $R > 3q > 4\kappa$, also $q^4 \mid X$. Set

$$w = X/q, \qquad B_t = (K_0 + w)C - F, \qquad B = qB_t.$$

The remaining partial fractions give $q^3 \mid Y$ and $q - 1 \mid B_t$, using $W = 1 \pmod{q-1}$. Thus $t = B_t/(q-1) \geq 0$ is an integer. The fixed-minus auxiliary parity lemma gives $R = 3 \pmod 4$, and the same ratio rounding recovers the source half-binomial Y.

The parent's fresh unscaled source reconstruction, at $A_{\text{source}} = Y(X+1)+2$, now applies to the recovered data. It does not reuse I_* or equate the two Pell parameters. The pinned source61/native/controller calibration yields a finite syntactic refutation. This proves soundness with exactly that inherited calibration.

9.5 Completeness by a new strong Pell period

A finite refutation supplies the parent's canonical source data with $X = 2^R > q$, $R = 3 \pmod 4$, $r \geq 48$, $q^3 \mid Y$, $q^4 \mid X$ and $t \geq 0$. Retain J, F, α, Y and set $x = X - q > 0$. The unchanged zero-offset ratio lemma supplies positive integers

$$c = \psi_A(R), \quad k = 2\psi_{2V_0+1}(r+1), \quad \tau = \chi_{2V_0+1}(r+1),$$

$$\eta = c - kY > 0, \qquad \zeta = k - \eta > 0, \qquad \rho = (k - R - 1)/\mu^2 \geq 1.$$

Choose a positive m such that $\tau c^2 \mid \psi_A(m)$. This exists by finiteness: the determinant-one recurrence matrix

$$\begin{pmatrix} 2A & -1 \\ 1 & 0 \end{pmatrix}$$

has a positive power equal to the identity modulo τc^2. Its corresponding Pell ordinate is zero modulo that modulus. Set

$$I_* = \psi_A(m), \qquad i = \frac{I_*}{\tau c^2} > 0, \qquad f = \chi_A(m).$$

Then $E = ic^2$, $D = \tau^2$ and $\mathcal{F} = 1 + \Delta DE^2 = f^2$. No divisibility of the parent's former strong ordinate is assumed.

The retained universal auxiliary-completeness lemma applies to this arbitrary new strong pair because $R = 3 \pmod 4$, $c = \psi_A(R)$ and $c \mid I_*$. It supplies positive T_*, u_* with

$$V = f^2(cT_* - R) - c, \qquad y = V + u_*, \qquad QV^2 - (Q-1)y^2 = 1.$$

The same explicit growth estimate gives $T_* > \alpha$; put $T = T_* - \alpha > 0$. Thus the auxiliary witness is also genuinely rebuilt.

Projection gives an integer $k_+ = K_+/H$ with $0 < k_+ < Xc$. As in the parent, $f > (A-1)I_* > Xc$ and $\tau > Y \geq Y/q^3$. Since $\tau, f \geq 3$, $\tau f > \tau + f > Y/q^3 + k_+$. Define

$$n = Wt + \rho + \tau f + u_* - Y/q^3 - k_+ + X/q^4.$$

It is a positive integer, including when $t = 0$. Substitution into (5) gives $Z/M = \tau f + y$, so one factor of the quartic norm vanishes. This constructs all ten positive coordinates from every refutation. A pointwise positive inverse between the old and new witness coordinates is neither needed nor asserted.

9.6 Exact degree and reproducible scope

Give each variable degree one and each fixed coefficient degree zero. The literal degree bounds include

$$A : 4,\ D : 16,\ E : 5,\ \mathcal{F} : 34,\ Q : 42,\ V : 38,\ L : 25,\ W : 27,\ M : 13,\ Z : 51.$$

The uncancelled syntactic bound on $\mathcal{N}$ is 288. Put $Z_0 = Z - MV$; its degree is 40, with highest part furnished uniquely by $-H\mu^2 q^3 WB$. The exact identity

$$\omega := s - v = \delta Z_0^2 + 2\delta MVZ_0 - M^2(V^2 - 1)$$

has summand bounds 122, 133 and 102. Also $\deg u = 118$ and $\deg v = 144$. Hence

$$\mathcal{N} = (\omega - u)^2 - 4uv$$

has unique top candidate ω^2 of degree 266; the other term bounds are 251, 236 and 262.

Set $q_0 = \beta J$, $X_0 = q_0 + x$, $k_0 = \eta + \zeta$ and

$$B_0 = (q_0 K_0 + X_0)(q_0 - F - \alpha) - q_0 F.$$

The complete leading form is

$$\omega_0 = 2^7 q_0^{63} X_0^{15} i^5 Y^{31} k_0^{16} B_0 (q_0 - F).$$

The highest power of x in ω_0^2 is 32. Selecting $\alpha^2 F^2 \eta^{32}$ in the remaining factors gives

$$[J^{126} x^{32} i^{10} Y^{62} \eta^{32} \alpha^2 F^2]\mathcal{N} = 2^{14}\beta^{126} > 0.$$

The exponents sum to 266, proving the exact total degree at the actual fixed tuple.

The portable packet contains the full circuit, coherent coefficient payloads, copied ordinary dependencies, `build.py`, and a Python standard-library-only independent `check.py`. Both normal and optimized Python runs check the complete literal circuit, all fixed payload hashes, the full norm and centering identities, 24 exact independent-formula controls, actual-payload modular controls, nonzero leading homogeneous witnesses, six finite Pell-period/fused-kernel cases, and five rejected mutations. These finite checks are transcription and algebra checks; they do not replace the unbounded proof or the inherited native/source calibration. The original construction sources are pinned to repository commit `84e16e43092614bae262ca59651cb9d77e31a255`.

10 A completed ordinary power graph with nine auxiliaries

Result and scope. For positive integers n, k and a natural m, the relation $m = n^k$ has the following ordinary polynomial graph with **nine auxiliary natural variables and exact degree** 20. More generally, r simultaneous positive-power requests use $4r + 5$ auxiliaries and exact degree $12r + 8$. The inputs are separate from the auxiliaries. This is a complete polynomial result, not a special-function interface, and does not claim a new fewest-variable consistency equation.

The common scale and common Pell-index certificate are inherited from the reviewed synchronized construction. The new reduction eliminates the common Pell base as an independent coordinate even though its substituted expression $A = B - Ub$ can be negative. Instead of introducing a sign witness, we prove that the entire negative branch is harmless. No historical minimality claim is made.

10.1 The explicit ordinary polynomial

Fix $r \geq 1$ and inputs $n_i, k_i \geq 1$, $m_i \geq 0$. Quantify only

$$z, U, v, q, b, \qquad y_i, h_i, s_i, t_i \quad (1 \leq i \leq r)$$

over $\mathbb{N}$. Define polynomial abbreviations

$$\begin{aligned}
&W = 2 + \sum_i (n_i + k_i + m_i), && Y_i = k_i + y_i, && D = \prod_i Y_i,\\
&V = D^2(v+1), && B = 1 + 4D(q+1), && A = B - Ub,\\
&T_i = 2An_i - n_i^2 - 1, && X_i = Y_i(A - n_i) + m_i + T_i h_i, &&\\
&S_i = X_i + Us_i, && R_i = k_i + 4Y_i t_i. &&
\end{aligned}$$

Set $N(a, x, y) = x^2 - (a^2 - 1)y^2 - 1$. The entire equation is

$$\boxed{\begin{aligned} E_r = \big[A^2 - W^3(W+2)(z+1)^2 - 1\big]^2 + N(A, U, V)^2 \\ + \sum_{i=1}^{r} \big(N(A, X_i, Y_i)^2 + N(B, S_i, R_i)^2\big) = 0. \end{aligned}} \tag{9}$$

All products have the fixed length r, and every exponent in the displayed recipe is the numeral 2 or 3. In particular A, T_i, X_i are polynomial abbreviations, not additional variables or assumed positive quantities.

Theorem. For the stated input domain, $E_r = 0$ has a solution in its $4r + 5$ natural auxiliaries if and only if $m_i = n_i^{k_i}$ for every i.

10.2 The signed elimination argument

Suppose $E_r = 0$. All residuals vanish. The scale equation yields $|A| > W^2$, while $Y_i \geq k_i \geq 1$ and $n_i, k_i, m_i \leq W$. The potentially dangerous case is $A = -\alpha < 0$. Put $\tau_i = 2\alpha n_i + n_i^2 + 1 > m_i$. Then

$$X_i = -(\alpha + n_i)Y_i + m_i - \tau_i h_i < 0.$$

Writing $x_i = -X_i > 0$, its norm gives $x_i \leq \alpha Y_i$. Its definition therefore implies

$$n_i Y_i + \tau_i h_i \leq m_i, \qquad Y_i \leq m_i \leq W < \alpha.$$

If $Y_i > 1$, the factorization

$$(\alpha Y_i - x_i)(\alpha Y_i + x_i) = Y_i^2 - 1$$

has positive integer factors, with the second greater than αY_i. Thus $Y_i^2 - 1 > \alpha Y_i$, contradicting $Y_i < \alpha$. Hence $Y_i = 1$, $k_i = 1$, $x_i = \alpha$, and $m_i = n_i + \tau_i h_i$. Since $\tau_i > m_i$, we obtain $h_i = 0$ and $m_i = n_i = n_i^{k_i}$.

This proves soundness of the complete negative branch without using an index theorem, changing a congruence sign, or adding a positivity witness. The branch is real: the checker includes explicit negative-A zeros. If any requested exponent is at least two, that branch is impossible.

10.3 Positive soundness

For $A > 0$ we have $A > W^2$, so $A \geq 2$, $A > n_i$, and

$$T_i \geq 2A - 2 > W \geq m_i.$$

All coordinates needed by the standard Pell-index criterion are now nonnegative, and $Y_i, V > 0$, $B \geq 5$. The definitions supply

$$Y_i^2 \mid V, \quad B \equiv 1 \pmod{4Y_i}, \quad B \equiv A \pmod{U}, \quad S_i \equiv X_i \pmod{U}, \quad R_i \equiv k_i \pmod{4Y_i}.$$

Together with the three norms and $k_i \leq Y_i$, the exact Matiyasevich criterion identifies $X_i = \chi_A(k_i)$ and $Y_i = \psi_A(k_i)$, where

$$\chi_a(j) + \psi_a(j)\sqrt{a^2 - 1} = (a + \sqrt{a^2 - 1})^j.$$

The imported criterion includes its signed-index analysis; it is the statement exposed by Mathlib's `Pell.matiyasevic` and the repository's `PellRepresentation.criterion_iff_pell`.

The scale norm is at Pell base $W + 1$, with second coordinate $W(z + 1) > 0$. Classification gives an index $j > 0$. Modulo W, its recurrence gives $\psi_{W+1}(j) \equiv j$, so $W \mid j$ and $j \geq W$. Consequently

$$A \geq (W + 1)^j > W^W \geq n_i^{k_i}, \qquad T_i > n_i^{k_i}.$$

For $H_j = \chi_A(j) - (A - n_i)\psi_A(j)$ the exact recurrence is

$$H_0 = 1, \qquad H_{j+1} = n_i H_j + T_i \psi_A(j).$$

Thus $H_{k_i} \equiv n_i^{k_i} \pmod{T_i}$. The definition of X_i also gives $H_{k_i} = m_i + T_i h_i$. Both proposed and true outputs lie in $[0, T_i)$, establishing equality.

10.4 Simultaneous completeness

Assume all requested powers are true, and calculate W first. Choose

$$A = \chi_{W+1}(W), \quad z = \psi_{W+1}(W)/W - 1, \quad X_i = \chi_A(k_i), \quad Y_i = \psi_A(k_i).$$

The recurrence modulo W makes z natural. Set $y_i = Y_i - k_i$ and

$$h_i = \sum_{j=0}^{k_i - 1} n_i^{k_i - 1 - j} \psi_A(j) \geq 0.$$

The recurrence above proves the required expression for X_i. These powers and finite sums occur in the witness proof only, not in equation (9).

Put $D = \prod_i Y_i$. In the finite ring

$$(\mathbb{Z}/4D^2\mathbb{Z})[s]/(s^2 - (A^2 - 1)),$$

the element $A + s$ is a unit with inverse $A - s$. A positive finite-group order supplies $U = \chi_A(\ell) \equiv 1 \pmod{4D^2}$ and $V = \psi_A(\ell) \equiv 0 \pmod{4D^2}$, with $V > 0$. Hence $v = V/D^2 - 1 \geq 0$, and $\gcd(U, 4D) = 1$. CRT gives

$$B \equiv A \pmod{U}, \qquad B \equiv 1 \pmod{4D}.$$

Increase B by a multiple of $4DU$ until $B > A$ and $B > 1$, and set

$$b = (B - A)/U, \qquad q = (B - 1)/(4D) - 1.$$

Finally take $S_i = \chi_B(k_i)$, $R_i = \psi_B(k_i)$ and

$$s_i = (S_i - X_i)/U, \qquad t_i = (R_i - k_i)/(4Y_i).$$

Polynomial dependence on the base proves these quotients integral. Monotonicity for $B > A$ proves them nonnegative. Every residual vanishes, and $A = B - Ub$ restores exactly the chosen scale. No source consistency, prime-distribution assertion, or circular existence argument is used.

10.5 Honest counts and the tiling boundary

With independent input ports,

$$\deg V = 2r + 1, \quad \deg A \le r + 1, \quad \deg X_i, \deg S_i \le r + 3.$$

The auxiliary norm has degree $6r + 4$, while each local norm has degree at most $2r + 6$. Its leading component contains $-16q^2v^2(\prod_i Y_i)^6$, so the final degree is exactly $12r+8$. For one request, the delivered DAG has 12 total variables, 9 auxiliaries, 4 residuals and degree 20. For two requests it has 19 total variables, 13 auxiliaries, 6 residuals and degree 32.

Retaining A as an independent natural variable and adding the residual $B - A - Ub$ gives a lower-degree sibling. For two requests this uses 14 auxiliaries and degree at most 24. This is a coordinate–degree tradeoff, not universal dominance.

The preceding two-power strict-boundary tiling interface has 8 exposed coordinates. Literal substitution now gives an ordinary polynomial with 21 natural variables, or 22 for the lower-degree sibling. For k tiles, the complete substituted degree bounds are respectively $1728k + 8$ and $1152k + 8$; they are *not* the standalone graph degrees 32 and 24. The complete four-corner fixture has bounds 6920 and 4616, including every fixed-power packed-circuit operation.

Not a new consistency endpoint. A strict-boundary tile set correctly initialized and calibrated to an actual ZFC proof checker is still required. The raw 943-tile list cannot simply be substituted: it contains an all-zero tile, admitting a trivial one-cell rectangle under this generic boundary condition. We therefore claim a completed power-graph lemma and generic tiling compiler, not a 21-variable Con(ZFC) polynomial or an improvement to the calibrated 9/10-positive and 11-integer frontiers.

Validation. The packet includes complete polynomial DAGs, separately transcribed evaluators, 1500 modular whole-polynomial comparisons, 16 full positive- and negative-base exponent-one witnesses, genuinely different-index CRT fixtures, 479080 finite negative-base controls, exact-degree certificates, and corruption tests. Normal and optimized Python runs pass using explicit exceptions rather than disabled assertions. Source hashes are pinned. These tests supplement the universal proof; no new Lean verification, encoded internal ZFC proof object, or proof of consistency is claimed.

Independent ordinary review. The separate review accepts the signed branch, positive branch, common-scale bound and simultaneous CRT construction. Its independent symbolic checker compares every generic port and residual with the literal emitted DAG and confirms exact degrees 20, 32 and 24 for the three generic artifacts. It does not certify the separate tiling-source calibration.

Sources and reproduction. The copied Mathlib source is pinned to commit `4dc43842a11f` (full hash in the source pin); its relevant Pell-index theorem is included with its license. The exact repository wrapper and inherited shared-frame proofs are separately authenticated in `SOURCE_PINS.json`. No incoming archive was changed.

The complete recipe can be rebuilt with `python build.py -check`; the separate checker is `python check.py`. Repeat both with `python -O` to verify that no crucial test relies on an assertion which optimized mode can remove. The numerical input-domain restrictions remain part of the theorem in every mode.

11 Fixed numeral definitions and reproducibility

The integer equations use one complete source34 tuple. Put $\rho = 2^{121}$ and $B = \rho^{8801} = 2^{1064921}$. The supplied file `integer/coefficients/recipe/wang943_tiles.json` is the entire fixed ordered list of

943 tiles, in west–north–east–south order. It is reconstructed, entry for entry, from all 430 transitions of the included fixed blank machine. The first two selectors are the specified start and forbidden tiles; their order is part of the numeral definition. No search for a halting computation enters this finite reconstruction.

For tile i let x_{it} be its four fixed labels. Set

$$\begin{aligned} W_j &= 2^{10+2j}, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960), \\ E &= \{0, \dots, 942\} \cup \{o_t + 2j : 0 \le t < 4,\ 0 \le j < 10\} \\ &\quad \cup \{1063, 1064, 1095, 1096, 2195, 3296\}, \\ c_i &= 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) W_{10t+j} + \sum_{j=40}^{43} W_j \quad (0 \le i \le 942). \end{aligned}$$

At a copy position put $c_{o_t+2j} = W_{10t+j}$; at the four anchors $1095, 1096, 2195, 3296$ put $W_{40}, W_{41}, W_{42}, W_{43}$ respectively. The two dummy coefficients at $1063, 1064$ are zero. Thus $|E| = 989$.

Define the fixed integers

$$\begin{aligned} \mu &= 1022 + \sum_{j=0}^{43} W_j, \\ Z_p &= (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}, \\ D_C &= \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e \in E} c_e (\rho^{4393-e} + \rho^{5495-e}), \\ M_C &= B - 1 - \sum_{e \in E,\ e \ne 1} \rho^e - 2\rho^{1063}, \\ M_{F,n} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) \\ &\quad + \rho^{2195} + \rho^{3296} + Z_p. \end{aligned}$$

The actual polynomial ports are

$$\beta = B - 1, \quad K_0 = D_C + B\rho, \quad M_C, \quad M_F = M_{F,n} + B - 1,$$

with the two additional cone numerals $A_c = 2^{1113}$ and $B_c = 2^{666104}$. The six complete payloads total 615,883 uncompressed bytes. A payload with little-endian bytes $b_0, \dots, b_{m-1}$ means precisely the integer $\sum_{i=0}^{m-1} b_i 256^i$, not a digest standing in for an unspecified number. Payload hashes, finite recipe reconstruction, complete tile equality, and the ordinary coefficient-adapter proof are all supplied.

The ten-positive construction deliberately uses a different complete compact44 tuple, with $\beta = 2^{590069} - 1$. Its four whole payloads total 295,388 bytes; its actual bytes and source-specific binding are retained under `positive/coefficients`, with the pinned ordinary adapter texts under `positive/source`. None of its ports is substituted into a source34 integer construction. In both tuples M_F has already been shifted; shifting it a second time would define the wrong polynomial.

12 Calibration and the meaning of the checks

The source34 integer chain uses the pinned ordinary signed-kernel theorem, its mixed-field coefficient adapter, initialized finite-strip extraction, and the exact K215/refutation correspondence. The source44 positive chain uses its own normalized positive kernel and compact compiler adapter. The

file `baseline/CALIBRATION_CITATIONS.md` identifies each theorem, source path, commit and available copied proof. These are different coherent coefficient interfaces, not interchangeable symbols satisfying only a few numerical bounds.

The new work proves arithmetic witness transformations on top of that ordinary chain. For a reader who has not accepted the inherited calibration, the independently stated new conclusions are predecessor zero-existence equivalences, together with the identified calibration obligations. This boundary is not repaired by running more test assignments, authenticating a digest, or citing a theorem about a related machine without its adapter.

The repository's external Lean theorem about the exact machine and its syntactic refutation predicate is also distinct from a newly exported object-language ZFC proof of each polynomial biconditional. No such new export is claimed here. This iteration does not resume the paused formalization project.

12.1 Reproduction

Each component includes its literal circuit, an independently transcribed specification or checker, ordinary proof and source bindings. Exact polynomial identities and nonzero leading coefficients provide algebraic and degree evidence. Finite Pell examples and deliberate corruptions test transcription and exceptional cases; they are not substitutes for the unbounded existence and recovery arguments.

The release-level reproduction script runs the listed checks in normal and optimized Python, authenticates the payloads, and checks that literal artifacts rebuild without byte changes. Its receipt distinguishes executed checks from inherited mathematical assumptions. The source ZIP contains this report's TeX and PDF, executable checks and the actual fixed integer payloads. It requires no solution of the resulting consistency equation and makes no search claim about whether such a solution exists.

12.2 What remains open

There is no single definition of "simplest" that makes all the rows comparable. Nine-variable equations retain a variable-count advantage; shorter operation-count formulas with more variables remain relevant; and direct quartics optimize degree instead. Further compression must preserve the unbounded reverse witness construction, rather than only produce a smaller circuit on a few sample assignments.

The generic tiling compiler in the separate exponentiation section is not silently instantiated with the raw 943-tile list. That list contains an all-zero tile. A bare strict-boundary rectangle-existence test would accept a trivial rectangle and would not express the required initialized computation. A valid new endpoint needs an explicit initialized source adapter, with its proof, before its variable count can be advertised as a Con(ZFC) construction.

References

[1] ProveIt contributors. Diophantine consistency research at the pinned comparison commit. Commit `84e16e430926`, 24 October 2026.

[2] ProveIt contributors. Scalar intersective carrier after signed main-root recovery. The four-root predecessor and its fixed-witness interpolation dependency.

[3] ProveIt contributors. Direct positive base with the zero-offset main parameter. The positive source reconstruction, ratio and whole-product projection lemmas used by the coprime-fused successor.

[4] ProveIt contributors. Inherited ordinary signed-kernel and native proof. Sections 3–7; exact local source hashes are recorded in the archive.

[5] ProveIt contributors. Source34 coefficient and compiler proof. The whole tuple, initialization convention and arithmetic interface are part of the theorem, not interchangeable parameters.

[6] ProveIt contributors. Ordinary exact-machine and syntactic-refutation calibration. This is an inherited dependency, not a newly checked machine proof.